ma4261_hw1

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1 Problem 1

Consider random variables X and Y such that Y is uniformly distributed over $\mathcal{Y} = \{0,1\}^b$. Let $\hat{Y}$ be an estimate of Y obtained by observing X. Show that

$$Pr(\hat{Y} \neq Y) \geq 1 - \frac{I(X;Y)+1}{b}$$

Now, suppose that $X_1, X_2, ..., X_n, Y$ are random variables such that $X_1, X_2, ..., X_n$ are mutually independent and Y is as before. For each $i \in [n]$, let $\hat{Y}_i$ denote an estimate of Y obtained from observing X_i. Show that

$$\max_{1\leq i\leq n} Pr(\hat{Y}_i \neq Y) \geq 1 - \frac{1}{n} - \frac{1}{b}$$

1.1 $P(\hat{Y} \neq Y)$

By Fano inequality, we have

$$\begin{aligned}
Pr(\hat{Y} \neq Y) &\geq \frac{H(Y|X)-1}{\log|\mathcal{Y}|} \\
&= \frac{H(Y) - I(X;Y) - 1}{\log|\mathcal{Y}|} \\
&= \frac{\log|\mathcal{Y}| - I(X;Y) - 1}{\log|\mathcal{Y}|} && (Y \text{ uniform}) \\
&= 1 - \frac{I(X;Y)+1}{\log|\mathcal{Y}|} \\
&= 1 - \frac{I(X;Y)+1}{b} && (|\mathcal{Y}| = 2^b)
\end{aligned}$$

1.2 $\max_{1\leq i\leq n} Pr(\hat{Y}_i \neq Y)$

By Fano inequality, for each $i \in [n]$

$$Pr(\hat{Y}_i \neq Y) \geq 1 - \frac{I(X_i;Y)}{b} - \frac{1}{b}$$

Then,

$$\max_{1\leq i\leq n} Pr(\hat{Y}_i \neq Y) \geq \frac{1}{n}\sum_{i=1}^{n} Pr(\hat{Y}_i \neq Y) \geq 1 - \frac{\sum_{i=1}^{n} I(X_i;Y)}{nb} - \frac{1}{b}$$

We will show that $-\frac{\sum_{i=1}^{n} I(X_i;Y)}{nb} \geq -\frac{1}{n}$, that is $\sum_{i=1}^{n} I(X_i;Y) \leq b$. Indeed, as $X_1, X_2, ..., X_n$ are independent

$$\sum_{i=1}^{n} I(X_i;Y) = I(X_1^n;Y)$$

Hence

$$\sum_{i=1}^{n} I(X_i;Y) = I(X_1^n;Y) = H(Y) - H(Y|X_1^n) \leq H(Y) = b$$

Lemma 1. *If X, Z are independent, then*

$$I(X,Z;Y) = I(X;Y) + (Z;Y)$$

Proof. $I(X,Z;Y) = H(X,Z) - H(X,Z|Y) = (H(X)+H(Z)) - (H(X|Y)+H(Z|Y)) = I(X;Y) + (Z;Y)$ □

2 Problem 2

Let $L_\epsilon(X)$ be defined as the minimum length of a source code designed for source $X \sim p_X$ with error probability ϵ, i.e.

$$L_\epsilon(X) = \inf\{\log M : \exists(1, M) - \text{source code with } P_e \leq \epsilon\}$$

1. Show that if there exists a λ such that $Pr(-\log p_X(X) \leq \lambda) \geq 1 - \epsilon$, then $L_\epsilon(X) \leq \lambda$
2. A common qualification of randomness, besides the entropy is the Rényi entropy of order $\alpha \in (0, 1) \cup (1, \infty)$ defined by

$$H_\alpha(X) = \frac{1}{1-\alpha} \log \sum_{x \in \mathcal{X}} p_X(x)^\alpha$$

   Show that $\lim_{\alpha \to 1} H_\alpha(X) = H(X)$
3. Show that for fixed p_X, the map $\alpha \mapsto H_\alpha(X)$ is non-increasing
4. Prove that for every $\alpha \in (0, 1)$ and $\epsilon > 0$

$$L_\epsilon(X) \leq H_\alpha(X) + \frac{1}{1-\alpha} \log \frac{1}{\epsilon} + 1$$

5. Prove that for every $\beta > 1$ and $\delta \in (0, 1 - \epsilon)$

$$L_\epsilon(X) \geq H_\beta(X) - \frac{1}{\beta - 1} \log \frac{1}{\delta} - \log \frac{1}{1 - \epsilon - \delta}$$

6. Show that if $X^n = (X_1, X_2, ..., X_n)$ consists of i.i.d random variables $X \sim p_X$, then $H_\alpha(X^n) = nH_\alpha(X)$ for all $\alpha > 0$
7. Use the above parts to prove fixed-to-fixed length source coding theorem with strong converse, i.e. you should provide bounds on

$$\liminf_{n \to \infty} \frac{1}{n} L_\epsilon(X^n) \text{ and } \limsup_{n \to \infty} \frac{1}{n} L_\epsilon(X^n)$$

2.1 if there exists a λ such that $Pr(-\log p_X(X) \leq \lambda) \geq 1 - \epsilon$, then $L_\epsilon(X) \leq \lambda$

Let

$$A_\lambda = \{x \in \mathcal{X} : -\log p_X(x) \leq \lambda\} \subseteq \mathcal{X}$$

Given $P(A_\lambda) \geq 1 - \epsilon$, we will upper bound the size of A_λ so that there is a code that map A_λ injectively into M code words.

If $x \in A_\lambda$, then

$$\begin{aligned} -\log p_X(x) &\leq \lambda \\ \log p_X(x) &\leq -\lambda \\ p_X(x) &\geq 2^{-\lambda} \end{aligned}$$

Moreover,

$$\begin{aligned} 1 &= \sum_{x \in \mathcal{X}} p_X(x) \\ &\geq \sum_{x \in A_\lambda} p_X(x) && (A_\lambda \subseteq \mathcal{X}) \\ &\geq \sum_{x \in A_\lambda} 2^{-\lambda} && (x \in A_\lambda \implies p_X(x) \geq 2^{-\lambda}) \\ &= 2^{-\lambda} |A_\lambda| \end{aligned}$$

Then, $|A_\lambda| \leq 2^\lambda$. Let $M = 2^\lambda$, there exists an injective map from A_λ into $[M] = \{1, ..., M\}$. As $P(A_\lambda) \geq 1 - \epsilon$, $P_e < \epsilon$, hence, $L_\epsilon(X) \leq \log M = \lambda$

2.2 $\lim_{\alpha\to 1} H_\alpha(X) = H(X)$

Let $f(\alpha) = \log \sum_{x\in\mathcal{X}} p_X(x)^\alpha$, f is analytic on $(0,\infty)$, we have (log denotes $\log_2$)

$$f'(\alpha) = \frac{1}{\sum_{x\in\mathcal{X}} p_X(x)^\alpha} \sum_{x\in\mathcal{X}} p_X(x)^\alpha \log p_X(x)$$

Let $g(\alpha) = \frac{1}{1-\alpha}$, g is analytic on $\mathbb{R}\setminus\{1\}$, we have

$$g'(\alpha) = \frac{1}{(1-\alpha)^2}$$

Write $\alpha \mapsto f(\alpha)$ as a Taylor series around $\alpha = 1$, and note that as f is analytic around 1, $f^{(n)}(1)$ is finite for all n, then

$$f(\alpha) = f(1) + f'(1)(\alpha-1) + o(\alpha-1)$$

where $\frac{o(\alpha-1)}{\alpha-1} \to 0$ as $\alpha \to 1$. Therefore, around $\alpha = 1$

$$\begin{aligned} H_\alpha(X) &= g(\alpha)f(\alpha) \\ &= \frac{1}{1-\alpha}(f(1) + f'(1)(\alpha-1) + o(\alpha-1)) \\ &= f(1) + f'(1) + \frac{o(\alpha-1)}{\alpha-1} \end{aligned}$$

Hence,

$$\lim_{\alpha\to 1} H_\alpha(X) = f(1) + f'(1) = H(X)$$

2.3 for fixed p_X, the map $\alpha \mapsto H_\alpha(X)$ is non-increasing

For any $\alpha \in (0,1)\cup(1,\infty)$,

$$\frac{d}{d\alpha} H_\alpha(X) = g'(\alpha)f(\alpha) + g(\alpha)f'(\alpha)$$

Note that $g'(\alpha) = \frac{1}{(1-\alpha)^2} > 0$, we will show that $-f(\alpha) - \frac{g(\alpha)}{g'(\alpha)} f'(\alpha) \geq 0$ for all $\alpha \in (0,1)\cup(1,\infty)$, let $s = \sum_{x\in\mathcal{X}} p_X(x)^\alpha$

$$\begin{aligned} -f(\alpha) - \frac{g(\alpha)}{g'(\alpha)} f'(\alpha) &= -\log s + (\alpha-1)\frac{\sum_{x\in\mathcal{X}} p(x)^\alpha \log p(x)}{s} \\ &= -\log s + \frac{1}{s}\sum_{x\in\mathcal{X}} (\alpha-1)p(x)^\alpha \log p(x) \\ &= \left(\frac{1}{s}\sum_{x\in\mathcal{X}} -p(x)^\alpha \log s\right) + \left(\frac{1}{s}\sum_{x\in\mathcal{X}} p(x)^\alpha \log p(x)^\alpha - p(x)^\alpha \log p(x)\right) \\ &= \frac{1}{s}\sum_{x\in\mathcal{X}} p(x)^\alpha[-(\log s) + \log p(x)^\alpha - \log p(x)] \\ &= \sum_{x\in\mathcal{X}} \frac{p(x)^\alpha}{s} \log \frac{p(x)^\alpha/s}{p(x)} \\ &= \sum_{x\in\mathcal{X}} q(x) \log \frac{q(x)}{p(x)} && \text{(where } q(x) = \frac{p(x)^\alpha}{s}\text{)} \\ &= D(q\|p) \geq 0 && (q \text{ is a distribution on } \mathcal{X}) \end{aligned}$$

2.4 for every $\alpha \in (0,1)$ and $\epsilon > 0$, $L_\epsilon(X) \leq H_\alpha(X) + \frac{1}{1-\alpha}\log\frac{1}{\epsilon} + 1$

Let

$$A_\alpha = \left\{x \in \mathcal{X} : -\log p(x) \leq H_\alpha(X) + \frac{1}{1-\alpha}\log\frac{1}{\epsilon}\right\}$$

If $x \notin A_\alpha$, then

$$\begin{aligned}
-\log p(x) &> H_\alpha(X) + \frac{1}{1-\alpha}\log\frac{1}{\epsilon} \\
\log p(x) &< -H_\alpha(X) + \frac{1}{1-\alpha}\log\epsilon \\
p(x) &< 2^{-H_\alpha(X)}\epsilon^{1/(1-\alpha)} \\
p(x)^{1-\alpha} &< \epsilon 2^{-(1-\alpha)H_\alpha(X)} && (1-\alpha > 0)
\end{aligned}$$

Then

$$\begin{aligned}
P(A_\alpha^C) &= \sum_{x\notin A_\alpha} p(x) \\
&= \sum_{x\notin A_\alpha} p(x)^{1-\alpha}p(x)^\alpha \\
&< \epsilon 2^{-(1-\alpha)H_\alpha(X)} \sum_{x\notin A_\alpha} p(x)^\alpha \\
&\le \epsilon 2^{-(1-\alpha)H_\alpha(X)} \sum_{x\in\mathcal{X}} p(x)^\alpha \\
&= \epsilon 2^{-(1-\alpha)H_\alpha(X)} 2^{(1-\alpha)H_\alpha(X)} = \epsilon
\end{aligned}$$

Hence, $P(A_\alpha) \ge 1-\epsilon$, from (1) we have $L_\epsilon(X) \le H_\alpha(X) + \frac{1}{1-\alpha}\log\frac{1}{\epsilon} < H_\alpha(X) + \frac{1}{1-\alpha}\log\frac{1}{\epsilon} + 1$.

2.5 for every $\beta > 1$ and $\delta \in (0, 1-\epsilon)$ $L_\epsilon(X) \ge H_\beta(X) - \frac{1}{\beta-1}\log\frac{1}{\delta} - \log\frac{1}{1-\epsilon-\delta}$

Lemma 2 (Han-Verdú)**.** *If there exists λ such that $Pr(-\log p(X) \ge \lambda) \ge 1-\delta$, then $L_\epsilon(X) \ge \lambda - \log\frac{1}{1-\epsilon-\delta}$*

Proof of Lemma **??**. Suppose, there is a code of length L with error probability $P_e \le \epsilon$. Let

$$\begin{aligned}
T &= \{x \in \mathcal{X} : -\log p(X) \ge \lambda\} = \{x \in \mathcal{X} : p(X) \le 2^{-\lambda}\} \\
S &= \{x \in \mathcal{X} : \phi f x = x\}
\end{aligned}$$

Then,

$$\begin{aligned}
P(T) &= P(T \cap S^C) + P(T \cap S) \\
&\le P(S^C) + P(T \cap S) \\
&= P_e + P(T \cap S) \\
&\le \epsilon + P(T \cap S) \\
&= \epsilon + \sum_{x\in T\cap S} p(x) \\
&\le \epsilon + 2^{-\lambda}|T \cap S| \\
&\le \epsilon + 2^{-\lambda}|S|
\end{aligned}$$

S is the set that is decoded correctly, so the map $f : S \to [2^L]$ is injective, hence $|S| \le 2^L$. Therefore,

$$1-\delta \le P(T) \le \epsilon + 2^{-\lambda}|S| \le \epsilon + 2^{-\lambda+L}$$

Then $L \ge \lambda - \log\frac{1}{1-\epsilon-\delta}$, hence

$$L_\epsilon(X) \ge \lambda - \log\frac{1}{1-\epsilon-\delta}$$

□

Main Proof. Let

$$B_\beta = \left\{x \in \mathcal{X} : -\log p(X) \ge H_\beta(X) - \frac{1}{\beta-1}\log\frac{1}{\delta}\right\}$$

If $x \notin B_\beta$, then

$$\begin{aligned}
-\log p(x) &< H_\beta(X) - \frac{1}{\beta-1}\log\frac{1}{\delta} \\
\log p(x) &> -H_\beta(X) + \frac{1}{1-\beta}\log\delta \\
p(x) &> 2^{-H_\beta(X)}\delta^{1/(1-\beta)} \\
p(x)^{1-\beta} &< \delta 2^{-(1-\beta)H_\beta(X)} && (1-\beta<0)
\end{aligned}$$

Then

$$\begin{aligned}
P(B_\beta^C) &= \sum_{x\notin B_\beta} p(x) \\
&= \sum_{x\notin B_\beta} p(x)^{1-\beta}p(x)^\beta \\
&< \delta 2^{-(1-\beta)H_\beta(X)} \sum_{x\notin B_\beta} p(x)^\beta \\
&\le \delta 2^{-(1-\beta)H_\beta(X)} \sum_{x\in\mathcal{X}} p(x)^\beta \\
&= \delta 2^{-(1-\beta)H_\beta(X)} 2^{(1-\beta)H_\beta(X)} = \delta
\end{aligned}$$

Hence, $P(B_\beta) \ge 1-\delta$, from Lemma **??**, we have $L_\epsilon(X) \le H_\beta(X) - \frac{1}{\beta-1}\log\frac{1}{\delta} - \log\frac{1}{1-\epsilon-\delta}$

□

2.6 if $X^n = (X_1, X_2, ..., X_n)$ consists of i.i.d random variables $X \sim p_X$, then $H_\alpha(X^n) = nH_\alpha(X)$ for all $\alpha > 0$

Lemma 3. *If X, Y are independent, then $H_\alpha(X,Y) = H_\alpha(X) + H_\alpha(Y)$*

Proof of Lemma **??**.

$$\begin{aligned}
H_\alpha(X,Y) &= \frac{1}{1-\alpha}\log\sum_{x\in\mathcal{X}}\sum_{y\in\mathcal{Y}} p_{X,Y}(x,y)^\alpha \\
&= \frac{1}{1-\alpha}\log\sum_{x\in\mathcal{X}}\sum_{y\in\mathcal{Y}} p_X(x)^\alpha p_Y(y)^\alpha && \text{(independent)} \\
&= \frac{1}{1-\alpha}\log\left[\left(\sum_{x\in\mathcal{X}} p_X(x)^\alpha\right)\left(\sum_{y\in\mathcal{Y}} p_Y(y)^\alpha\right)\right] \\
&= \frac{1}{1-\alpha}\left[\log\left(\sum_{x\in\mathcal{X}} p_X(x)^\alpha\right) + \log\left(\sum_{y\in\mathcal{Y}} p_Y(y)^\alpha\right)\right] \\
&= H_\alpha(X) + H_\alpha(Y)
\end{aligned}$$

□

Main Proof. Generalize from the case of two variables □

2.7 prove the fixed-to-fixed-length source coding theorem with strong converse

Fix $0 < \alpha < 1 < \beta$, ϵ, δ, we have

$$H_\beta(X) + \frac{1}{n}\left(-\frac{1}{\beta-1}\log\frac{1}{\delta} - \log\frac{1}{1-\epsilon-\delta}\right) \le \frac{1}{n}L_\epsilon(X^n) \le H_\alpha(X) + \frac{1}{n}\left(\frac{1}{1-\alpha}\log\frac{1}{\epsilon} + 1\right)$$

Therefore,

$$H_\beta(X) \le \liminf_{n\to\infty}\frac{1}{n}L_\epsilon(X^n) \le \limsup_{n\to\infty}\frac{1}{n}L_\epsilon(X^n) \le H_\alpha(X)$$

As $H_\beta(X) \to H(X)$ as $\beta \to 1^+$ and $H_\alpha(X) \to H(X)$ as $\alpha \to 1^-$, therefore $\lim_{n\to\infty} \frac{1}{n} L_\epsilon(X^n)$ exists and

$$\lim_{n\to\infty} \frac{1}{n} L_\epsilon(X^n) = H(X)$$

for all $\epsilon \in (0,1)$ (strong converse)

3 Problem 3

An $n \times n$ non-negative matrix $W = [W_{ij}]$ is called doubly stochastic if $\sum_i W_{ij} = 1$ and for all j and $\sum_j W_{ij} = 1$ for all i

1. Let $p = (p_1, p_2, ..., p_n)$ be a probability vector, i.e. $p_i \geq 0$ for all i and $\sum_i p_i = 1$. Let $q = pW$ where W is doubly stochastic. Show that q is a probability vector and $H(q) \geq H(p)$.
2. Show that stationary distribution μ for a doubly stochastic matrix W is the uniform distribution
3. Conversely, prove that if uniform distribution is a stationary distribution for a Markov transition matrix W, then W is doubly stochastic.

3.1 q is a probability vector and $H(q) \geq H(p)$

Entries of W and p are non-negative, hence entries of q are non-negative. Let $1 = (1, 1, ..., 1) \in \mathbb{R}^n$ be a row vector. We will show that $q1^T = 1$, that is the sum of entries of q is 1.

$$\begin{aligned} q1^T &= pW1^T \\ &= p1^T && \text{(sum of each row of } W \text{ is 1)} \\ &= 1 && \text{(sum of entries of } p \text{ is 1)} \end{aligned}$$

3.2 stationary distribution μ for a doubly stochastic matrix W is the uniform distribution

As sum of each column of W is 1, $1W = 1$, therefore, $\mu = \left(\frac{1}{n}, \frac{1}{n}, ..., \frac{1}{n}\right)$ being a uniform distribution is a left eigen vector of W, i.e. a stationary distribution

3.3 if uniform distribution is a stationary distribution for a Markov transition matrix W, then W is doubly stochastic

Markov transition matrix has sum of each row is 1, furthermore, $\mu = \left(\frac{1}{n}, \frac{1}{n}, ..., \frac{1}{n}\right)$ is stationary distribution, i.e. a left eigen vector of eigen value 1. Hence, $1W = 1$, that is equivalent to sum of each column of W is 1, therefore, W is doubly stochastic

4 Problem 4

Consider a discrete memoryless source X with alphabet $\{1, 2, .., , M\}$. Suppose that the symbol probabilities are ordered and satisfy $p_1 > p_2 > ... > p_M$ and also satisfy $p_1 < p_{M-1} + p_M$. Let $l_1, l_2, ..., l_M$ be the length of prefix-free code of minimum expected length for such a source.

1. TRUE or FALSE: $l_1 \leq l_2 \leq ... \leq l_M$. Provide a short justification
2. Show that if Huffman algorithm is used to generate the above code, then $l_M \leq l_1 + 1$
3. Show that $l_M \leq l_1 + 1$ for any (not necessarily Huffman generated) prefix-free code of minimum expected length.
4. Suppose $M = 2^k$ for some integer k. Show that all codewords must have the same length.

4.1 TRUE or FALSE: $l_1 \leq l_2 \leq ... \leq l_M$

TRUE

Suppose the otherwise, $i < j$ and $l_i > l_j$, as $p_i > p_j$, we have $p_i l_i + p_j l_j > p_i l_j + p_j l_i$, therefore, if we swap the code words of i and j, we will have a **strictly** lower expected length, that contradicts the minimum expected length.

4.2 if Huffman algorithm is used to generate the above code, then $l_M \leq l_1 + 1$

if Huffman algorithm is used, $l_M = l_{M-1}$ as code of M and $M-1$ are siblings. In Huffman algorithm, M and $M-1$ is merged into a new symbol with probabiltiy $p_M + p_{M-1}$. Let the merged symbol be 0 with probability $p_0 = P_M + P_{M-1}$. As $p_0 > p_1$, in the reduced problem of alphabet $\{0, 1, 2, ..., M-2\}$, we have $p_0 > p_1$, therefore, $l_0 \leq l_1$. As $l_M = l_{M-1} = l_0 + 1$, therefore, $l_M = l_0 + 1 \leq l_1 + 1$

4.3 Show that $l_M \leq l_1 + 1$ for any (not necessarily Huffman generated) prefix-free code of minimum expected length

Let c_1, c_M, c_{M-1} be the code words for $1, M, M-1$.

In minimum expected length code, the code words for M and $M-1$ must be siblings, that is c_M and c_{M-1} differ by only the last bit. Let c be the common prefix of c_M and c_{M-1}. We construct a new code

$$\begin{aligned} c'_1 &= c \\ c'_M &= c_1 \oplus 0 \\ c'_{M-1} &= c_1 \oplus 1 \\ c'_i &= c_i && \text{(if } i \neq 1, M, M-1) \end{aligned}$$

That is, the new code for 1 is the common prefix of the original code of $M, M-1$, the new code for M is the original code for 1 appended by the bit 0, the new code for $M-1$ is the original code for 1 appended by the bit 0. As the original code is optimal, we must have

$$\begin{aligned} p_1 l_1 + p_M l_M + p_{M-1} l_{M-1} &\leq p_1(l_M - 1) + p_M(l_1 + 1) + p_{M-1}(l_1 + 1) \\ (p_M + p_{M-1} - p_1) l_M &\leq (p_M + p_{M-1} - p_1) l_1 + (p_M + p_{M-1} - p_1) \\ l_M &\leq l_1 + 1 \end{aligned}$$

4.4 Show that all codewords must have the same length

We have

$$l_1 \leq l_2 \leq ... \leq l_M \leq l_1 + 1$$

Kraft inequality

$$1 \geq \sum_{i=1}^{M} 2^{-l_i} \geq \sum_{i=1}^{M} 2^{-l_M} = M 2^{-l_M} = 2^{k - l_M}$$

Therefore, $l_M \geq k$

Case 1: $l_M = k$

Let $[M] = I \amalg J$ such that $l_i = k - 1$ for all $i \in I$ and $l_j = k$ for all $j \in J$. Note that $J \neq \varnothing$ because $M \in J$. Now, we construct a new code as follows:

$$\begin{aligned} c'_i &= c_i \oplus 0 && \text{(for all } i \in I) \\ c'_j &= c_j && \text{(for all } j \in J) \end{aligned}$$

That is, for each $i \in I$, we construct a new code word c'_i by appending c_i with the bit 0. The new code lies totally at the k-th level of the tree. There are $M = 2^k$ nodes at the k-th level, therefore, the map from new code words to the k-th level is a bijection. If $I \neq \varnothing$, let $i \in I$, the binary string $c_i \oplus 1$ at the k-th level is not in $\{c_j : j \in J\}$ due to $\{c_i : i \in [M]\}$ is prefix-free and it is not in $\{c'_i : i \in I\}$ because it ends with 1. Therefore, $I = \varnothing$, that is, all code words must have the same length.

Case 2: $l_M \geq k + 1$

Let $[M] = I \amalg J$ such that $l_i = l_M - 1 \geq k$ for all $i \in I$ and $l_j = l_M$ for all $j \in J$. Note that $J \neq \varnothing$. We construct a new code by a bijection from $[M] = 2^k$ to all nodes at the k-th level. The new code expected length is k, the original code expected length is

$$\sum_{i=1}^{M} p_i l_i = \sum_{i \in I} p_i (l_M - 1) + \sum_{j \in J} p_j l_M \geq \sum_{i \in I} p_i k + \sum_{j \in J} p_j l_M > \sum_{i \in I} p_i k + \sum_{j \in J} p_j k = k$$

The strict inequality yields a contradiction on optimality of the original code.

Remark 1. *The same reasoning can be used for the case $l_M < k$, however, both cases $l_M < k$ and $l_M = k$ use the same method as in the proof of Kraft inequality.*

5 Problem 5: Random coding for Huffman codes

Consider a binary prefix-free (PF) code with code word lengths

$$l_1 \leq l_2 \leq ... \leq l_M$$

We construct this PF randomly as follows: For each $k \in [M]$ the code word $C(k)$ of length l_k is chosen independently from the set of all 2^{l_k} possible binary strings with length l_k according to the uniform distribution. Let $P_M(good)$ be the probability that the so-constructed code if PF.

1. Consider a source with binary alphabet so $M = 2$ and there are only two lengths $l_1 \leq l_2$. Show that

$$P_2(good) = (1 - 2^{-l_1})^+$$

   where $x^+ = \max(0, x)$

2. Prove by induction on M that

$$P_M(good) = \prod_{k=1}^{M} \left(1 - \sum_{j=1}^{k-1} 2^{-l_j}\right)^+$$

3. Is the following statement true or false. Provide a brief reason:

   $P_M(good) > 0$ *if and only if there exists a PF code for a source with alphabet size* M

4. Use the above parts to prove Kraft inequality.

5.1 $P_2(good)$

Let $\{C(i) \sim C(j)\}$ denote the event where $\{C(i), C(j)\}$ has a common prefix. Then,

$$\begin{aligned}
1 - P_2(good) &= P(\{C(1) \sim C(2)\}) \\
&= Pr(C(1)_1 = C(2)_1, C(1)_2 = C(2)_2, ..., C(1)_{l_1} = C(2)_{l_1}) \\
&= Pr(C(1)_1 = C(2)_1) Pr(C(1)_2 = C(2)_2) ... Pr(C(1)_{l_1} = C(2)_{l_1}) \\
&= Pr(C(1)_1 = C(2)_1)^{l_1} \\
&= 2^{-l_1}
\end{aligned}$$

Then,

$$P_2(good) = 1 - 2^{-l_1} = (1 - 2^{-l_1})^+$$

5.2 $P_M(good)$

Suppose we already have the PF code words for $M - 1$ symbols of length $l_1, l_2, ..., l_{M-1}$. Consider a full binary tree of depth l_M, each code word of $1 \leq i \leq M - 1$ occupies some leaves of the tree, i.e. all leaves under that node. Sampling $C(M)$ so that the code remains PF is the same as picking a leave such that it is not occupied.

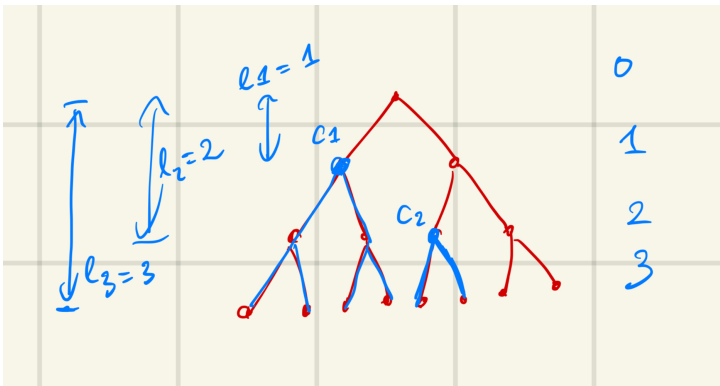

Figure 1: binary tree

The code $C(i)$ of length l_i occupies $2^{l_M - l_i}$ leaves, as $M-1$ code words is PF, then number of occupied leaves is

$$\sum_{i=1}^{M-1} 2^{l_M - l_i}$$

Uniformly sample $C(M)$ is equivalent to uniformly pick from 2^{l_M} leaves, then probability of $C(M)$ being PF with the rest is

$$\frac{2^{l_M} - \sum_{i=1}^{M-1} 2^{l_M - l_i}}{2^{l_M}} = 1 - \sum_{i=1}^{M-1} 2^{-l_i}$$

Let E_M be the event the first M code words being PF, we have

$$\frac{P(E_M)}{P(E_{M-1})} = \frac{P(E_M \cap E_{M-1})}{P(E_{M-1})} = P(E_M | E_{M-1}) = 1 - \sum_{i=1}^{M-1} 2^{-l_i}$$

By induction, we have

$$P_M(good) = P(E_M) = \prod_{k=1}^{M} \left(1 - \sum_{i=1}^{k-1} 2^{-l_i}\right)$$

This is true only for $P(E_i) > 0$ for all $1 \leq i \leq M-1$. When there exists $P(E_i) = 0$, let $N = \min_i \{i \in [M] : P(E_i) = 0\}$, the recurrence relation is true for all $1 \leq k \leq N$

$$\frac{P(E_k)}{P(E_{k-1})} = 1 - \sum_{i=1}^{k-1} 2^{-l_i}$$

and after that for all $N+1 \leq k \leq M$

$$1 - \sum_{i=1}^{k-1} 2^{-l_i} < 0$$

Hence, we can rewrite $P_M(good)$ by

$$P_M(good) = P(E_M) = \prod_{k=1}^{M} \left(1 - \sum_{i=1}^{k-1} 2^{-l_i}\right)^+$$

5.3 $P_M(good) > 0$ if and only if there exists a PF code for a source with alphabet size M

TRUE

As M and $l_1, l_2, ..., l_M$ are finite, total number of bits N is **finite**, hence if there exists a PF code, probability of sampling that code is $P_M(good) \geq 2^{-N} > 0$. On the other hands, $P_M(good)$ is the sum of all good code probabilities, $P_M(good) > 0$ implies there exists a good code. The situation is different if the sample space is not finite.

5.4 prove Kraft inequality

$P_M(good) > 0$ implies $\left(1 - \sum_{j=1}^{k-1} 2^{-l_j}\right)^+ > 0$ for all k. Take $k = M$, then

$$1 - \sum_{j=1}^{M-1} 2^{-l_j} > 0$$

Now we increase the size of input to $2^n M$ and the new lengths so that there are 2^n copies of l_i for each i, then there exists a PF code such that each old code word is replicated into 2^n copies, i.e

$$C(i) \mapsto b \oplus C(i)$$

for every $b \in \{0,1\}^n$. The new code is prefix-free and we have

$$1 - \frac{2^n - 1}{2^n} \sum_{j=1}^{M} 2^{-l_j} - \frac{1}{2^n} \sum_{j=1}^{M-1} 2^{-l_j} > 0$$

Taking limit at $n \to \infty$ recovers the Kraft inequality.